Predicting Future Uncontrolled Diabetes from Electronic Health Records

SASS

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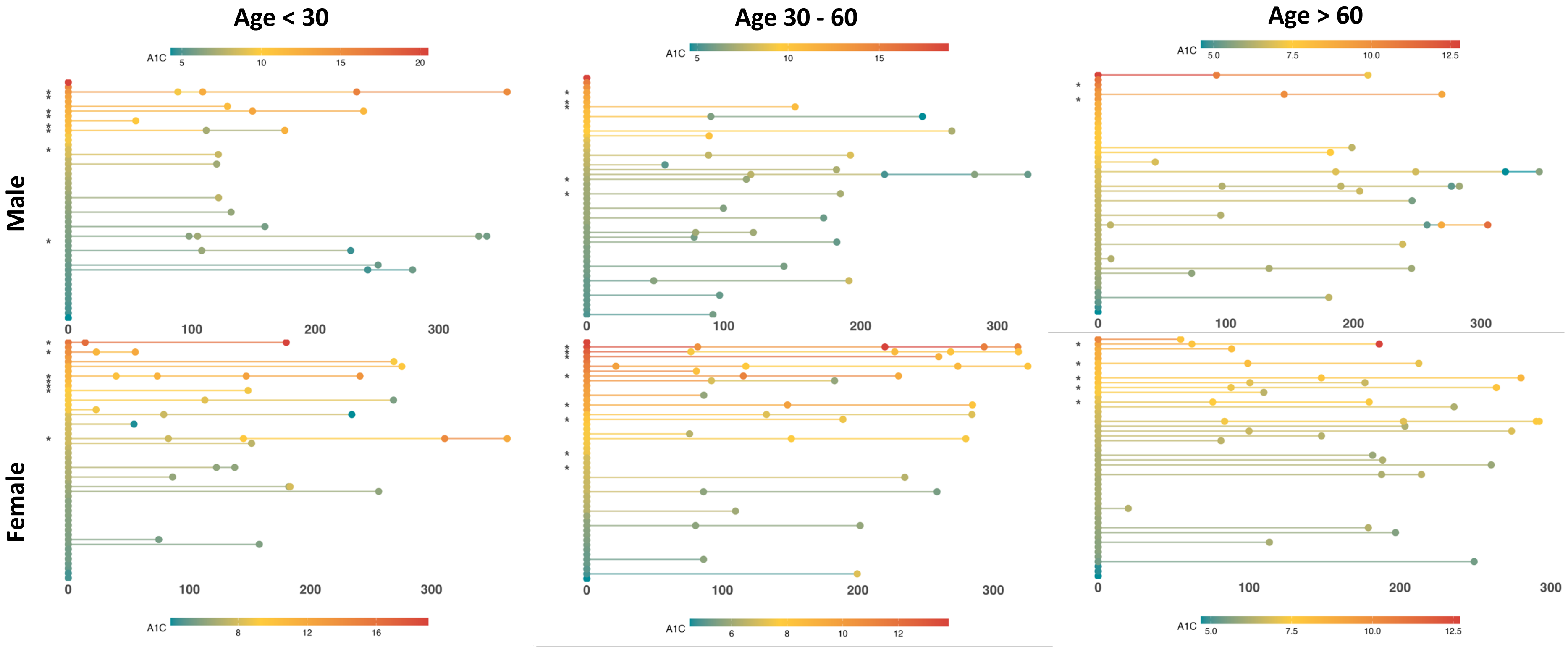
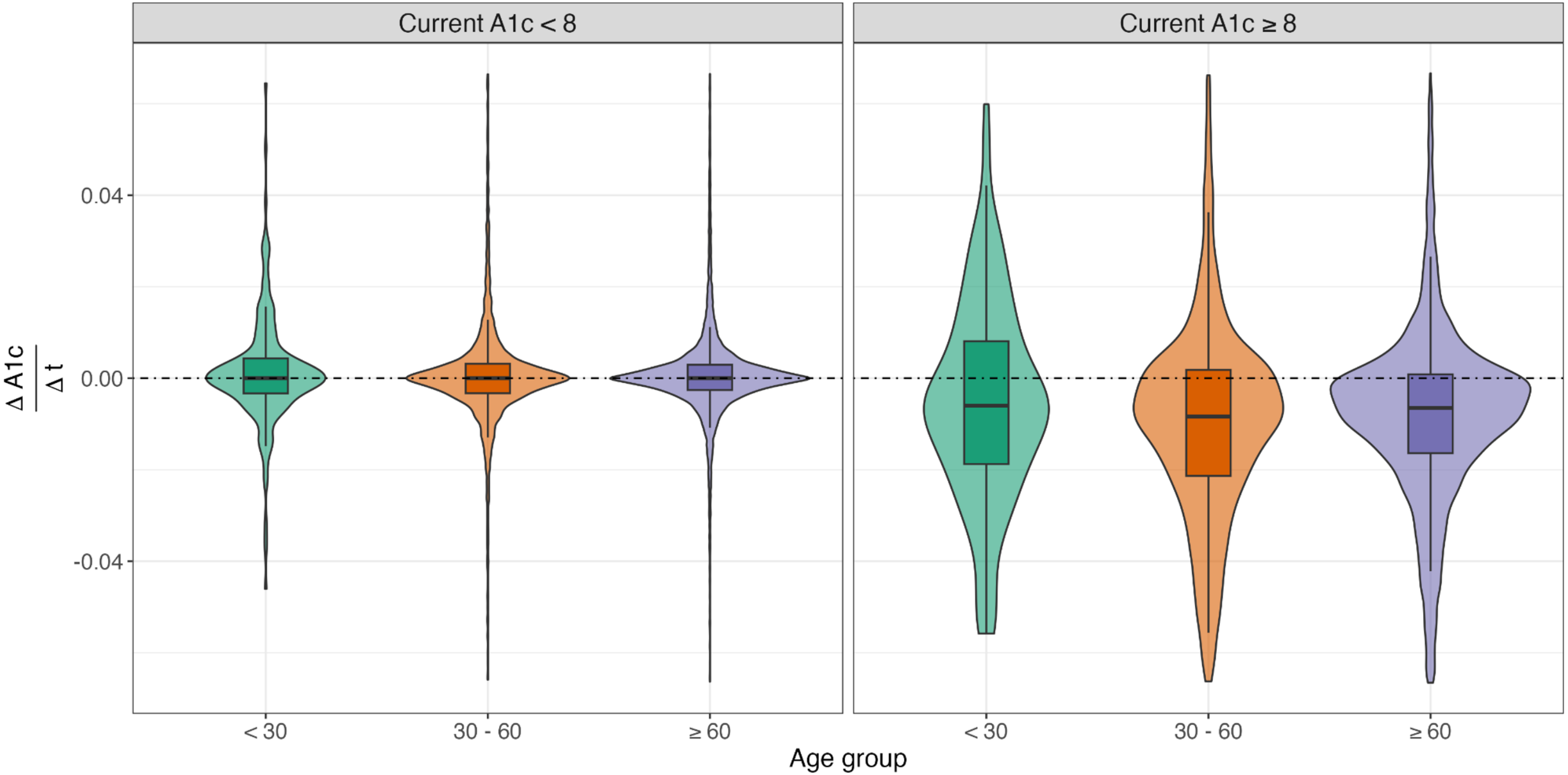
Landscape of the Dataset: Challenges and Patterns



x-axis denotes the number of days from the baseline

Distribution of Change Rate in A1c Measurements



Statistical Modelling Framework

Models

- Generalized Linear Model
- Generalized Additive Model
- Decision Tree
- Support Vector Machine
- Bayesian Hierarchical Model
- Random Forest
- 2-state transition modelling via Deep Neural Network
- Boosting

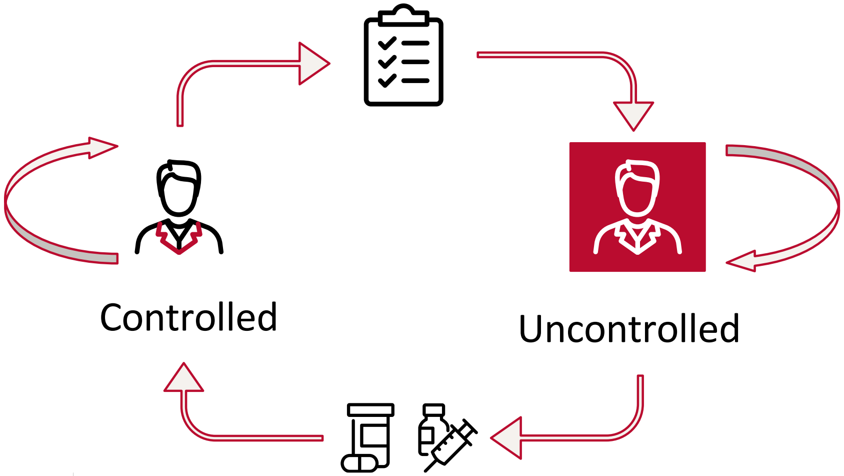
Missing Data Handling

- Mean / Median Imputation
- Bayesian Conditional Likelihood-based Imputation
- MICE (Multivariate Imputation by Chained Equations)

Imbalanced Data Handling

- Oversampling and undersampling.
- Class weighted loss (likelihood) optimization.
- Penalization
- F1-score optimization and cross-validation tuning.

2-state Transition Modeling via Shared Representations



$$P(Y_{it} = 1 \mid Y_{is} = 0) = \frac{\exp\{g_0(X_i, t - s)\}}{1 + \exp\{g_0(X_i, t - s)\}}$$

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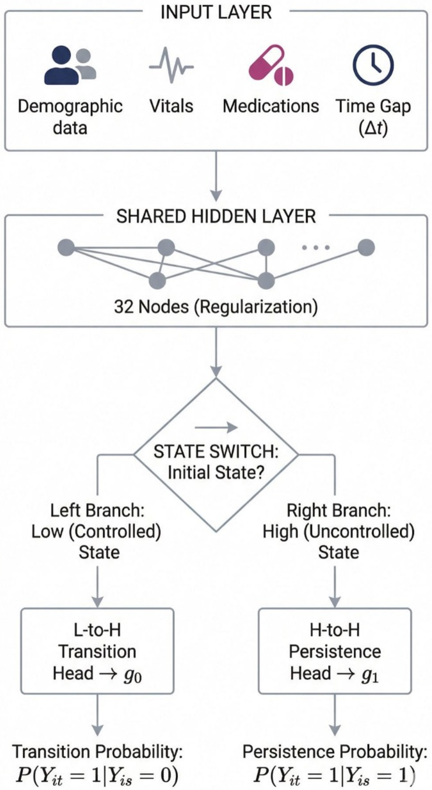
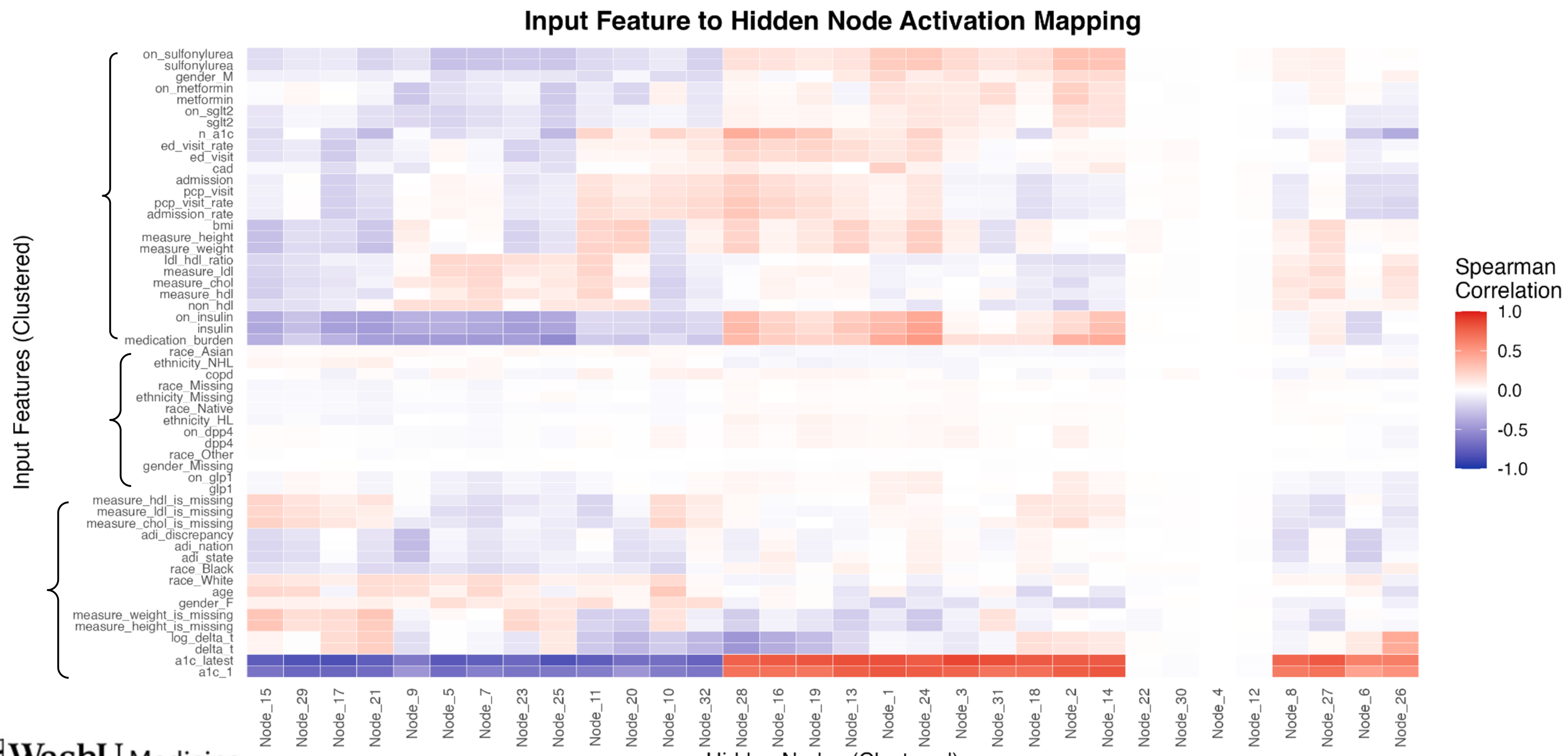


Hunting for Features through Data

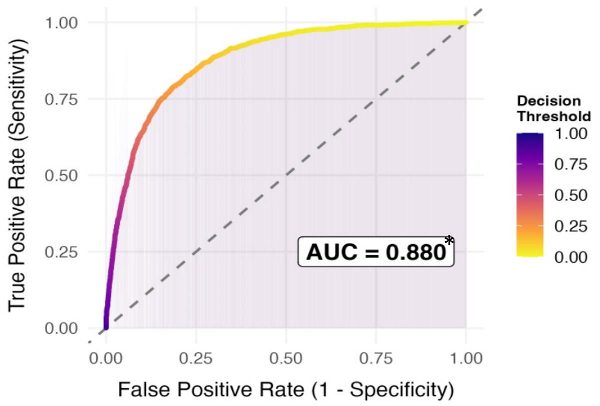


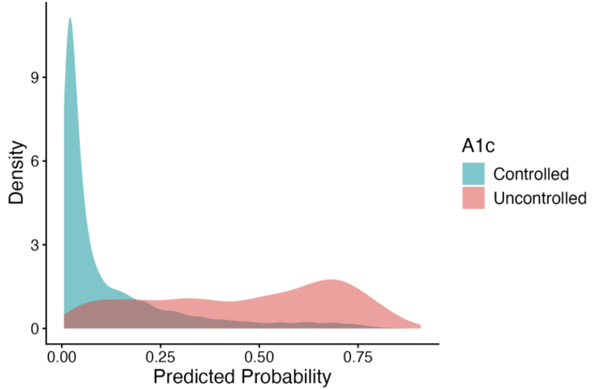
Biological Feature Hunting

Recency-weighted average of A1c	***A1c summary*** – smooths out the spikes**, weighted** toward the **present**
First A1c reading	Captures ***baseline severity*** independent of current treatment
Fraction of readings above 8	How often the patient crosses the uncontrolled threshold – ***chronic uncontrolled*** or ***transient spikes***?
A1c per treatment load	**Treatment-resistant**: high means uncontrolled A1c despite treatment
Days since last A1c	Long time gap between visits means reading is **outdated**: Consistent with ADA's recommendation of 3-6 month monitoring frequency
Biggest single-visit worsening	A sudden jump might suggest a **different clinical story** than a slow drift
Drift rate while controlled	For patients below uncontrolled threshold, A1c **can increase, stay flat, or decrease between visits**
Number of A1c readings	**More readings** suggest **unhealthy or more engaged patient**, and more **precise** estimates.

Predictive Power of XGBoost

Rank	Feature	Gain
1	**Days** since **last A1c** reading	**35.80%**
2	**Recency-weighted** average of past **A1c**	**20.40%**
3	Most **recent A1c**	**8.40%**
4	**First A1c** reading	**6.20%**
5	**Fraction** of readings **above 8.0**	4.70%
6	A1c **per treatment load**	3.20%
7	Biggest **single-visit worsening**	2.30%
8	A1c × drug-class count	2.20%
9	**HDL** cholesterol	1.90%
10	**Drift rate** while controlled	1.50%
11	**LDL** cholesterol	1.50%
12	Neighborhood deprivation (National **ADI**)	1.40%
13	Total **cholesterol**	1.40%
14	**Insulin** orders **per week**	1.30%
15	Patient **age**	1.30%





* AUC = **0.938** based on the organizers' test data

"All models are wrong" – So What Should We Do?

- Real EHR data is messy — out-of-order timestamps, some implausible values.
- We built our pipeline to carefully handle these issues.
- Our models learn what's in the chart, but reach a systematic ceiling.
- ***What it misses is what the patient does between visits*** – and that's exactly the information that would make it a practically useful decision tool.

What's not in the charts
Medication adherence
Diet and nutrition
Physical activities
Treatment timing (start/stop)
Financial and life disruptions
Mental health and stress

THANK YOU!

Questions?

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